



**Progressive Overload in Long-Term Exercise Interventions
Targeting Executive Function: A Scoping Review**

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Abstract

Executive function (EF), including inhibitory control, working memory, and cognitive flexibility, is essential for goal-directed behaviour across the lifespan. Regular exercise improves EF, with benefits moderated by frequency, intensity, time, and type. Progressive overload—systematic increases in exercise demands—may further optimize EF outcomes, yet its application in EF interventions remains unclear. The purpose of this review was to explore how progressive overload has been implemented in exercise interventions targeting EF. PRISMA-ScR processes were used to systematically search, select, and chart data from 44 studies published between 2014 and 2024. Findings were synthesized using quantitative summaries and qualitative thematic analysis. Most studies involved children, adolescents, or older adults. Moderate to large effect sizes were more commonly reported in interventions that progressively overloaded intensity and complexity, as well as in studies specifically targeting working memory and inhibitory control. Reporting inconsistencies and limited experimental manipulation remain key gaps. Future research should systematically manipulate progressive overload and explore occupational and resistance training contexts.

Keywords: Inhibitory control, working memory, cognitive flexibility, cognitive performance, training.

Beyond FITT: A Scoping Review on the Application of Progressive Overload in Long-Term Exercise Interventions Targeting Improvements in Executive Function

Executive function (EF) is an umbrella term encompassing cognitive processes that enable individuals to focus attention, resist impulses, and adapt to changing circumstances in pursuit of goal-directed behavior (Barkley, 2012; Diamond, 2013). EF is essential across the lifespan and in diverse contexts. In childhood, when EF is rapidly developing, it is foundational to academic success, particularly in mathematics and language (Spiegel et al., 2021). In adulthood, EF is associated with healthy behaviours (Gray-Burrows et al., 2019) and social functioning (Hilton et al., 2024). In older adults, as EF typically declines, it serves as a marker for maintaining independence (Verreckt et al., 2022).

EF is commonly described in terms of three interrelated but distinct core domains: inhibitory control, working memory, and cognitive flexibility (Diamond, 2013). Inhibitory control enables individuals to suppress automatic or impulsive responses in favor of deliberate, goal-oriented actions—for example, focusing on a conversation at a noisy party while ignoring background chatter. Working memory refers to the ability to hold and manipulate information over short periods, such as remembering a phone number long enough to dial it. Cognitive flexibility involves the capacity to shift perspectives and adapt to changing demands, such as rerouting a trip after encountering a road closure (Diamond, 2013). These core domains are often key outcomes in cognitive research and are typically assessed using standardized psychometric tools (Faria et al., 2015). Common examples include the Stroop Color-Word Test for inhibitory control (Faria et al., 2015; Golden, 1978), the Wechsler Adult Intelligence Scale (WAIS) Digit Span Test for working memory (Faria et al., 2015; Hill et al., 2010), and the Wisconsin Card Sorting Test for cognitive flexibility (Faria et al., 2015; Miles et al., 2021).

EF develops gradually through childhood and adolescence (Diamond, 2013), typically reaching full maturation in early adulthood (Tervo-Clemmens et al., 2023). While environmental influences such as nurturing caregivers (Bernier et al., 2010; Lynch et al., 2024), supportive schools (Cumming et al., 2020; Sankalaite et al., 2021), and rich learning environments (e.g., Diamond & Lee, 2011; Ernst et al., 2022) are critical for EF development, accumulating evidence from experimental studies indicates that

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regular exercise can play a significant role in enhancing EF across the lifespan (Singh et al., 2025). In children and adolescents, regular exercise has been associated with small to moderate improvements in EF, particularly in inhibitory control and working memory, when compared to sedentary or low-active peers (Haverkamp et al., 2020; Xue et al., 2019). In young and middle-aged adults, despite developmental peaks in EF, exercise continues to provide cognitive benefits, particularly enhancing inhibitory control and working memory. Small to moderate effect sizes have been observed across various exercise modalities when compared to sedentary or low-activity controls (Ludyga et al., 2023). In older adults, exercise appears to be protective against age-related declines in EF. In healthy populations, small to moderate effects have been reported—particularly for working memory—when compared to non-exercise or routine activity controls (Contreras-Osorio et al., 2022). Among individuals with mild cognitive impairment or dementia, moderate to large improvements in EF have been observed following physical exercise interventions, particularly at moderate intensity, compared to standard care or low-intensity control conditions (Chen et al., 2024).

The benefits of exercise on EF are moderated by the frequency, intensity, time, and type (FITT) of the activity. Moderate-to-vigorous exercise, performed three to four times per week for at least 30 minutes, yields significant improvements in EF compared to no exercise or lower-intensity conditions (Hsieh et al., 2021; Shi et al., 2022; Wang et al., 2023; Xue et al., 2019). Complex movements performed in unpredictable environments—such as in competitive sports—are most effective (Ludyga et al., 2020; Shi et al., 2022), particularly when combined with cognitively engaging tasks (Kolovelonis et al., 2022; Kolovelonis & Goudas, 2023; Mao et al., 2024). However, while much is known about how FITT parameters moderate exercise's impact on EF (e.g., Ludyga et al., 2020; Shi et al., 2022; Wang et al., 2023), less attention has been given to the dynamic adjustments of these parameters over time and their influence on long-term EF improvements.

The principle of progressive overload posits that gradual increases in exercise demands—whether through modifications of frequency, intensity, time or type—are necessary to elicit continued adaptation (Kraemer et al., 2020; Plotkin et al., 2022). In this context, we use the term exercise to encompass all

forms of structured physical activity and sport that are designed to promote adaptation and enhance EF. Progressive overload is typically defined by the target of progression, the frequency of progression, and the magnitude of progression (French, 2016; Gibson et al., 2024; Sheppard & Triplett, 2016). Investigating how progressive overload can optimize exercise-derived benefits is critical for designing long-term exercise interventions that bring about meaningful improvements in EF. Accordingly, the purpose of this scoping review was to explore how progressive overload has been implemented in long-term exercise interventions targeting EF. Specifically, it examined how progressive overload was implemented in these interventions and what, if any, impact it had on EF outcomes.

Methods

This scoping review was performed and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for scoping reviews (PRISMA-ScR) (Tricco et al., 2018). The process involved several tasks, including defining eligibility criteria, identifying studies, selecting relevant studies, charting data, and synthesizing the results.

Eligibility Criteria

To be included in this scoping review, studies had to meet specific criteria. They had to include at least one core domain of EF as an outcome measure and involve long-term (≥ 4 weeks) physical exercise interventions. Only peer-reviewed journal articles published between 2014 and 2024, written in English, and involving human participants were considered. Grey literature sources, such as conference abstracts and sources not discussing intervention details, were excluded. Additionally, cross-sectional studies and those involving acute exercise conditions were not included, as progressive overload could not have been applied in these cases.

Information Sources

The identification of potential studies involved searches in the databases PubMed, SPORTDiscus, and Web of Science, covering the period from January 2014 to December 2024. The search strategies were initially drafted by the primary investigator (first author) and refined through discussions with two additional reviewers (second and third author). The final PubMed search query was:

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1 (“executive function”[Title/Abstract] OR “inhibitory control”[Title/Abstract] OR “working
2 memory”[Title/Abstract] OR “cognitive flexibility”[Title/Abstract]) AND
3 (“exercise”[Title/Abstract] OR “physical activity”[Title/Abstract] OR “sport”[Title/Abstract])

4 The search results were exported into Zotero (Version 7.0; Fairfax, VA, USA) where duplicates were
5 removed by the primary investigator. Additional relevant studies were identified from the reference lists
6 of the retrieved studies.

7 Selection of Sources

8 To ensure consistency among authors, all retrieved studies were collaboratively reviewed by the authors.
9 The three authors worked in pairs of reviewers to evaluate the titles, abstracts, and full texts of all
10 retrieved studies. The focus was on identifying long-term intervention studies that incorporated
11 progressive overload. Disagreements regarding study selection and data extraction were resolved through
12 consensus and, when necessary, discussions with the third reviewer.

13 Data Charting

14 An initial data-charting form was developed by the primary investigator and discussed with the other
15 reviewers to determine the variables to be extracted. The three reviewers independently charted the data,
16 discussed the results, and continuously updated the data-charting form in an iterative process to ensure
17 clarity and completeness.

18 Data Items

19 The variables extracted from the included studies encompassed the source (authors), population, study
20 design, intervention parameters, progressive overload parameters, core domains of EF, and findings.
21 Study design was determined based on its alignment with the criteria for experimental, quasi-
22 experimental, or pre-experimental or meta-analytic research (Kowalski et al., 2022). Studies were
23 categorized as experimental if they included both a control group and randomization. Quasi-experimental
24 studies featured a control or comparison group but did not incorporate randomization (e.g., wait-list
25 designs). Pre-experimental studies lacked a control group and instead relied solely on pre- and post-test

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1 comparisons. Meta-analytic studies synthesized findings from multiple primary studies (Kowalski et al.,
2 2022).

3 Intervention parameters were extracted using the FITT principle—frequency, intensity, time, and
4 type. Progressive overload was operationalized across three dimensions: (1) the target of progression, (2)
5 the frequency of progression, and (3) the magnitude of progression. The target of progression
6 corresponded to components of the FITT principle that was systematically and intentionally increased. A
7 study was considered to have overloaded frequency if it increased the number of weekly exercise sessions
8 over time. Intensity was defined as an increase in exercise workload, typically measured using percentage
9 of heart rate maximum (%HRM), heart rate reserve (%HRR), or one repetition maximum (%1RM). Time
10 referred to an increase in session duration (e.g., progressing from 30 to 60 minutes). The type parameter
11 was broader and included any deliberate progression in the nature of the exercise—such as shifting from
12 cycling to running, or increasing difficulty or complexity, such as moving from basic to more cognitively
13 demanding tasks. The frequency of progression denoted the total number of overloads implemented over
14 the course of the intervention, while the magnitude of progression referred to the degree of change
15 introduced with each overload (e.g., a 10% increase in target heart rate percentage).

16 EF measures were categorized into the core domains, with the article by Diamond (2013) used as
17 a reference for grouping related constructs. Terms and measures encountered during the charting process
18 that differed from inhibitory control, working memory, and cognitive flexibility were categorized
19 accordingly, acknowledging the sometimes inconsistent use of terminology across the literature. For
20 instance, terms related to set shifting were charted under cognitive flexibility, while those referring to
21 updating were classified under working memory. Studies that assessed multiple EF components without
22 isolating specific core domains, or those that reported only composite EF scores, were charted as overall
23 EF. The findings extracted from each study were limited to the reported effects on core domains,
24 categorized as either a positive effect or no effect. If effect sizes were reported, their magnitude was also
25 charted in the findings column.

26 **Synthesis of Results**

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This scoping review mapped the extent to which studies reported progressive overload parameters in exercise interventions. Specifically, the review assessed whether studies provided details on all three key criteria of progressive overload: target of progression, the frequency of progression, and the magnitude of progression. Studies were categorized based on the completeness of their reporting, distinguishing those that included all three criteria from those that reported only one or two. To further explore trends in the literature, the target of progression (i.e., frequency, intensity, time, or type) was documented, and the number of studies reporting moderate to large effect sizes across these categories was recorded. This approach enabled an assessment of whether specific targets of progression were more commonly associated with greater reported effects. Effect size thresholds were defined as follows: small, η^2 or $\eta^2 \geq 0.01$, Cohen's d , Hedges' g , or standardized mean difference (SMD) ≥ 0.20 ; moderate, η^2 or $\eta^2 \geq 0.06$, d , g , or SMD ≥ 0.50 ; and large, η^2 or $\eta^2 \geq 0.14$, d , g , or SMD ≥ 0.80 (Cohen, 1988; Pallant, 2020). Additionally, studies were systematically grouped based on their experimental design, year of publication, population studied, and core domains of EF assessed. This categorization facilitated a broader mapping of the literature, providing insight into study characteristics and methodological trends.

Results

Selection of Sources for Evidence

A total of 1,335 studies were identified through searches in PubMed, SPORTDiscus, and Web of Science. Of these, 842 studies were removed because they were either duplicates ($n=36$), published prior to January 2014 ($n=158$), not written in English ($n=14$), or classified as grey literature sources ($n=634$). Grey sources, such as conference abstracts, newsletters, blogs, and other non-peer-reviewed online content, were excluded primarily due to their lack of sufficient methodological details regarding intervention specifics.

The remaining 493 studies were screened for relevancy, resulting in the exclusion of 333 studies based on title. The majority of these excluded studies were cross-sectional or acute in nature. Studies were excluded if their titles included terms such as cross-sectional ($n = 19$), relationship ($n = 22$), association ($n = 36$), correlation ($n = 2$), acute ($n = 117$), single-bout ($n = 12$), short-term ($n = 9$), brief ($n = 2$), post-

exercise (n = 4), or other terms indicating no long-term intervention (n = 54). Studies in which exercise, physical activity, or sport was not an independent variable were also excluded (n = 56).

From this process, 160 studies were sought for retrieval, but one study could not be accessed either through the institutional library or via interlibrary loan. This left 159 studies for eligibility assessment. During the eligibility assessment, an additional 22 studies were identified from reference lists and analyzed, bringing the total number of studies assessed for eligibility to 181. Of the 181 studies assessed, 137 were deemed not eligible on the basis that they did not report on progressive overload (n = 128), were animal studies (n = 4), were only protocols with no results (n = 2), were duplicates (n = 2), or were only abstracts with no information on progressive overload (n = 1). A total of 44 studies were included in the final scoping review. The full process of selecting sources for evidence is summarized in the PRISMA-ScR flow diagram (Figure 1).

[INSERT FIGURE 1 HERE]

Characteristics and Results of Sources of Evidence

Sources included randomized controlled trials, quasi-experimental studies, pre-experimental studies, and one systematic review. Populations ranged from preschool children to older adults, including athletic, clinical, at-risk, healthy, and neurodevelopmentally impaired groups. Interventions varied in frequency, intensity, time, and type. Core domains assessed included inhibitory control, working memory, and cognitive flexibility. See Table 1 for full details.

[INSERT TABLE 1 HERE]

Synthesis of Results

Of the 181 studies included for full-text review, 44 studies (24.3%) sufficiently addressed progressive overload. Publication frequency was grouped by period from January 2014 to December 2024 (see Figure 2): nine studies were published between 2014 and 2016, nine between 2017 and 2018, seven between 2019 and 2020, five between 2021 and 2022, and 14 between 2023 and 2024. In terms of study design (see Figure 3), the majority employed experimental methods (n=32), while others used quasi-experimental (n=8), pre-experimental (n=3), or meta-analytic approaches (n=1).

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1 [INSERT FIGURE 2 AND 3 HERE]

2 Regarding the population studied (see Figure 4), 15 studies targeted children and adolescents
3 aged 3 to 17 years, eight studies focused on adults aged 18 to 54 years, and 28 studies examined older
4 adults aged 55 years and above. To further contextualize the populations investigated, studies were
5 grouped by participant characteristics (see Figure 5). Healthy populations (n=21) included individuals
6 without diagnosed cognitive, developmental, or medical conditions. Clinical populations (n=9) involved
7 participants with diagnosed conditions such as mild cognitive impairment, Alzheimer's disease,
8 Parkinson's disease, coronary artery disease, and sleep apnea. At-risk populations (n=6) included
9 individuals with elevated risk factors such as obesity, physical inactivity, or substance dependence.
10 Neurodevelopmentally impaired populations (n=6) included children diagnosed with attention deficit
11 hyperactivity disorder, autism spectrum disorder, fetal alcohol spectrum disorder, or cerebral palsy.
12 Athletic populations (n=2) referred to individuals studied in the context of organized sport participation.

13 [INSERT FIGURE 4 AND 5 HERE]

14 Most studies reported positive intervention outcomes, with 38 studies demonstrating a beneficial
15 effect on EF, and six studies reporting no effect. No studies reported a negative effect. Among studies
16 reporting moderate to large effect sizes, one applied progressive overload through frequency, seven
17 through intensity, one through time, and 12 through type (typically via increased task complexity or
18 difficulty, see Figure 6). The number of studies reporting small, moderate, and large effect sizes was
19 pooled across core domains of EF (see Figure 7). Small effect sizes were most commonly reported in
20 studies where the intervention outcome was cognitive flexibility (n = 4), followed by working memory (n
21 = 3) and inhibitory control (n = 1). Moderate effect sizes were most commonly reported in studies where
22 the intervention outcome was working memory (n = 6), followed by cognitive flexibility (n = 4) and
23 inhibitory control (n = 3). Large effect sizes were most commonly reported in studies where the
24 intervention outcome was working memory (n = 5), followed by inhibitory control (n = 4) and cognitive
25 flexibility (n = 1). Finally, 15 studies fully reported on all three progressive overload parameters (target of

progression, the frequency of progression, and the magnitude of progression), while the remaining 29 studies provided only partial reporting, addressing one or two of these components.

[INSERT FIGURE 6 AND 7 HERE]

Discussion

Our synthesis identified three themes across the literature. First, the majority of studies focused on older adults, children, and adolescents as target populations. Second, exercise interventions that incorporated progressive overload consistently demonstrated positive effects on EF when compared to control groups or pre-intervention baselines, with the most notable gains associated with interventions that progressively overloaded intensity and type over time. The strongest domain-specific effects were observed in working memory and inhibitory control. Third, inconsistent reporting of progressive overload parameters, the absence of studies that examined progressive overload as an independent variable, and limited research applying progressive overload within occupational and resistance training contexts represent the main gaps in the literature.

Focus on Younger and Older Populations

When examining research involving children, the majority of studies focused on typically developing populations (e.g., Koutsandréou et al., 2016); however, several investigated children with neurodevelopmental impairments, including those with attention deficit hyperactivity disorder (e.g., Lee et al., 2017), fetal alcohol spectrum disorder (e.g., Pritchard Orr et al., 2018), cerebral palsy (e.g., Abo-Zaid et al., 2024), and autism spectrum disorder (e.g., Zhao et al., 2024). This distribution aligns with existing evidence suggesting that the cognitive benefits of exercise are most pronounced during key developmental periods in childhood, particularly among individuals exhibiting cognitive deficits (Singh et al., 2025). In neurodevelopmental impaired children, the potential for improvement may be greater, as exercise may serve to enhance underdeveloped cognitive capacities.

In applying progressive overload, the majority studies involving children and adolescents increased the complexity of activities to create greater challenges. For instance, Shen et al. (2020) gradually made dance tasks more demanding by moving from simple to more intricate routines,

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1 introducing new movement combinations, faster rhythms, and spatial formation changes. This stepwise
2 increase in difficulty placed ongoing cognitive and physical demands on participants, encouraging them
3 to adapt, remember, and switch between movements.

4 The small number of adult studies ($n = 8$) included in this scoping review focused primarily on
5 clinical or at-risk groups—particularly those where EF may be compromised by sleep, metabolic, or
6 cardiovascular dysfunction. For example, Kubitz et al. (2023) examined a 6-week aerobic and resistance
7 training program in overweight adults with obstructive sleep apnea, reporting a large improvement in
8 inhibitory control. Similarly, Peven et al. (2020) conducted a 12-month behavioral weight loss
9 intervention in adults with overweight and obesity, incorporating diet and varying levels of exercise.
10 However, despite the longer duration, no significant effects were observed for inhibitory control, working
11 memory, or cognitive flexibility.

12 In another clinical example, Vieira et al. (2018) conducted a 6-month randomized controlled trial
13 in men with coronary artery disease, comparing virtual reality exercise to paper-based programs and a
14 control. They found significant improvements in inhibitory control. Wang et al. (2017) examined a 12-
15 week aerobic training intervention in young adults with methamphetamine dependence, finding
16 reductions in drug cravings alongside improvements in both behavioral and neural measures of inhibitory
17 control. These findings underscore the cognitive relevance of exercise in populations facing vascular or
18 substance-related impairments in EF.

19 A common feature across these adult-targeted interventions was the application of progressive
20 overload through intensity. For instance, Wang et al. (2017) increased aerobic intensity from 65 to 70%
21 and then to 70 to 75% of maximum heart rate over 12 weeks. Vieira et al. (2018) used a staged
22 progression from 65% to 70% of heart rate reserve halfway through the program. Kubitz et al. (2023)
23 prescribed aerobic and resistance training at 60 to 90% of heart rate reserve, with resistance intensity
24 increasing from an initial 60% of 1-repetition maximum, though the precise frequency and magnitude of
25 overload were not specified.

1 In older adults, the benefits of exercise for EF appear particularly salient, as physical activity may
2 help protect against age-related cognitive decline (Liu et al., 2025). Accordingly, it is perhaps
3 unsurprising that the majority of progressive overload studies ($n = 28$) included in this review focused on
4 individuals aged 55 years and older. While several studies within this age bracket involved healthy older
5 adults, there was a noticeable trend toward prioritizing those experiencing accelerated cognitive decline.
6 This mirrors approaches used in child and adolescent populations with neurodevelopmental impairments,
7 where exercise interventions are often justified on the basis that such individuals stand to gain the most.
8 In older adults with neurodegenerative conditions, exercise similarly appears to offer its greatest cognitive
9 benefits among those at elevated risk. For example, Liu et al. (2022) demonstrated that an 8-week square-
10 stepping exercise program significantly improved working memory in individuals with Parkinson’s
11 disease. Similarly, Yu et al. (2018) implemented a 6-month aerobic cycling intervention in community-
12 dwelling older adults with mild to moderate Alzheimer’s disease, targeting cognitive flexibility and
13 inhibitory control. Although no significant improvements were observed, the authors noted that the
14 decline in EF was smaller than expected, suggesting a potential protective effect of aerobic exercise in
15 this population.

16 As with the adult cohort, progressive overload in older adults was most frequently applied
17 through gradual increases in exercise intensity. For example, Jonasson et al. (2016) implemented a six-
18 month aerobic training program for older adults that included walking, cycling, and cross-training.
19 Intensity was progressively increased from 40% to 80% of each participant’s estimated maximum heart
20 rate, with heart rate monitored throughout each session to ensure adherence. Similarly, Vidoni et al.
21 (2015) conducted a 26-week trial in which older adults engaged in moderate-intensity aerobic exercise—
22 primarily treadmill walking—with structured increases in heart rate reserve zones: 40 to 55% in Weeks 1
23 through 4, 50 to 65% in Weeks 5 through 18, and 60 to 75% in Weeks 19 through 26.

24 **Robust General and Specific Effects**

25 Of the 44 studies that incorporated progressive overload, 38 reported a positive effect, which suggests a
26 consistent benefit of structured, progressively overloaded exercise on EF. For example, Norouzi et al.

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(2019) implemented a 4-week motor-cognitive dual-task training program. EF was measured using a computer-based N-back task, and results showed a large effect on working memory. The authors attributed this effect to the dual-task nature of the intervention, which likely increased interaction between motor and cognitive brain networks, enhancing EF.

In contrast, six studies reported no effect. One such study, previously mentioned (Peven et al., 2020), examined the impact of a 12-month behavioral weight loss program, which included both moderate and high exercise conditions, on EF using the N-back, Task Switching, and Stroop tests. Despite the extended duration of the intervention, no significant improvements were observed on these three psychometric measures. However, the study did report a significant improvement in reward sensitivity, as measured by the Iowa Gambling Task, in the diet plus high-exercise group. The authors suggested that the lack of broader cognitive effects may have been due to low statistical power resulting from small intervention group sample sizes.

Importantly, the benefits of progressive overload have been demonstrated across a variety of populations and exercise types. For example, Biino et al. (2021) reported cognitive gains in children, while Wang et al. (2017) and Ghaleni et al (2024) found similar improvements in adults and older adults, respectively. These effects have also been observed across different training modalities, including aerobic exercise (Abo-Zaid et al., 2024), resistance training (Brydges et al., 2020), and exergaming (Hou et al., 2023).

While there appears to be a robust overall effect of progressive overload exercise on EF, a closer examination reveals that moderate to large effect sizes are most commonly reported in studies that progressively overloaded intensity and type. For example, Brydges et al. (2020) progressively overloaded intensity by increasing participants' target heart rate from 40% to 70 to 80% over the first 12 weeks of a 26-week walking intervention. Intensity was monitored using heart rate and ratings of perceived exertion, with the aim of enhancing both cardiorespiratory fitness and cognitive outcomes in older women with mild cognitive impairment. Similarly, Zhang et al. (2022) used a structured three-phase approach to gradually increase the cognitive demands of movement games over a 12-week program. Children began

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1 with simple motor games, then progressed to games that added cognitive rules (e.g., remembering
2 sequences, adapting to changes), and finally engaged in more challenging tasks combining complex
3 motor actions with dynamic, multi-step instructions.

4 Both intensity and type have been identified as key moderators of cognitive outcomes in
5 systematic reviews and meta-analyses. In terms of intensity, moderate levels are thought to optimize
6 arousal and facilitate neuroplasticity, while higher intensities trigger both acute physiological responses
7 (e.g., increased cerebral oxygenation) and longer-term adaptations (e.g., elevated BDNF), all of which
8 contribute to improvements in EF (Hsieh et al., 2021; Wang et al., 2023). With respect to type, evidence
9 suggests that movement complexity is a critical component of exercise interventions targeting EF.
10 Proposed mechanisms include heightened cognitive engagement and the activation of overlapping neural
11 pathways that support both motor control and EF (Ludyga et al., 2020; Shi et al., 2022).

12 It is also important to note that progressive overload interventions do not appear uniformly
13 effective across core domains of EF (see Figure 7). For inhibitory control and working memory, a
14 relatively greater number of studies report large effects, particularly for working memory, where large
15 effects were most frequently observed. In contrast, cognitive flexibility showed more small and moderate
16 effects, and the fewest large effects ($n = 1$) compared to inhibitory control and working memory. Fewer
17 observed large effects for cognitive flexibility may be attributable to several factors. First, only 20 studies
18 in our scoping review examined exercise's effects on cognitive flexibility, compared to 30 on working
19 memory, leaving cognitive flexibility underrepresented. Second, cognitive flexibility engages a broader
20 and more distributed neural architecture—including the prefrontal, cingulate, and parietal cortices (Dajani
21 & Uddin, 2015)—and is dependent on inhibitory control and working memory processes (Davidson et al.,
22 2006; Diamond, 2013; Garon et al., 2008). This broader neural integration may require a higher threshold
23 of stimulation to elicit meaningful improvements, potentially contributing to the attenuated effect sizes
24 observed for cognitive flexibility. Third, there may be an interaction between the developmental
25 trajectory of cognitive flexibility and the age focus of studies in our review. Cognitive flexibility matures
26 later than inhibitory control and working memory (Davidson et al., 2006; Diamond, 2013; Garon et al.,

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2008) and demonstrates greater stability or even improved efficiency with age, particularly in tasks requiring set-shifting (Hsieh & Chen, 2023). Consequently, in younger populations, cognitive flexibility may not yet be fully developed, limiting the observable benefits of exercise, while in older adults, ceiling effects may constrain further gains.

Reporting Limitations and Research Gaps

To systematically assess the effectiveness of progressive overload, it is essential that all three key parameters are clearly reported: (1) target of progression (i.e., frequency, intensity, time, type), (2) the frequency of progression (e.g., every week, every two weeks), and (3) the magnitude of progression (e.g., 5%, 10%). For instance, Hu et al. (2024) specified the target of progression—motor task difficulty (type)—but did not report its frequency or magnitude. Their intervention increased motor task difficulty based on participant mastery and incorporated cognitively demanding scenarios, while maintaining moderate exercise intensity through heart rate monitoring. Antunes et al. (2015) reported on both the target (time) and the magnitude of progression (progression from 20 to 60 minutes over six months), but omitted the overload frequency. In contrast, Iuliano et al. (2015) reported all three parameters. In their resistance training group, intensity increased every four weeks (from 60–70% to 80–85% 1-RM), and in the cardiovascular group, heart rate reserve targets increased every two weeks (from 50–60% to 70–80%), with frequency and session duration held constant.

Of the 44 studies reviewed, only 34% reported all three progressive overload parameters, which highlights a significant limitation in exercise intervention reporting. Because the effectiveness of progressive overload is inherently tied to its systematic application over time, incomplete reporting across studies hinders the ability to determine the specific role of progressive overload in moderating the effects of exercise on EF over time. It makes it difficult to replicate studies and may contribute to heterogeneity of findings across studies. Reporting standards such as the Preferred Reporting Items for Resistance Exercise Studies (PRIRES) and the Consensus on Exercise Reporting Template (CERT) emphasize the need to document progressive overload. While these guidelines require descriptions of the decision process behind progressive overload and how it was implemented, they may lack the specificity necessary

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1 to ensure consistent reporting of all three progressive overload parameters (Lin et al., 2023; Slade et al.,
2 2016). Future studies should explicitly report the target of progression, the frequency of progression, and
3 the magnitude of progression to enhance replicability and comparability.

4 Another notable gap in the literature is that none of the studies included in this scoping review
5 directly examined the effect of progressive overload on EF, with progressive overload explicitly treated as
6 the independent variable. This presents an opportunity for future research. Systematically manipulating
7 each progressive overload parameter—target, frequency, and magnitude—could yield important insights
8 into how exercise can be optimized to enhance cognitive outcomes. For example, a three-arm randomized
9 controlled trial could compare a non-progressive control group with two intervention groups that differ in
10 progression frequency (e.g., every two weeks vs. every four weeks) or progression magnitude (e.g., 10%
11 vs. 20% increases). Additionally, future studies could compare the cognitive effects of different targets of
12 progression, such as progression in intensity versus progression in task complexity.

13 Resistance training has been shown to improve EF (e.g., Brydges et al., 2020), yet a key gap
14 identified in this scoping review is the lack of resistance training interventions. Of the 44 studies
15 reviewed, only six included resistance training, and just two examined it in isolation (Brydges et al.,
16 2020; Iuliano et al., 2015). The remaining four combined resistance training with aerobic (Hu et al., 2024;
17 Kubitz et al., 2023; Mora-Gonzalez et al., 2023) or aerobic and balance training (Wang et al., 2018). This
18 limited focus on isolated resistance training presents a valuable opportunity for future research to clarify
19 its unique effects on EF. For example, studies could compare cognitive outcomes across a control group,
20 a power-based compound movement group, and a single-joint isolation exercise group, with each
21 condition matched for progressive overload. Such research would help clarify how resistance training—
22 specific parameters (e.g., neural drive, compound versus isolation exercises) influence EF.

23 Lastly, there appears to be an abundance of intervention studies designed to support development
24 of EF in younger populations and slow their decline in the elderly; however, little attention has been given
25 to the benefits such interventions might confer in occupational contexts. Public safety professions such as
26 firefighting, paramedicine, and policing place substantial demands on EF due to the combined effects of

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1 physical exertion, prolonged heat exposure, and/or psychological stress. These factors have been shown to
2 impair core domains including working memory, inhibitory control, and cognitive flexibility, resulting in
3 more decision-making errors (Coehoorn et al., 2020) and slower reaction times (Thompson et al., 2025).
4 EF is critical for safety, operational effectiveness, and rapid decision making in the field—often with lives
5 at stake. Research is needed to test and implement exercise-based interventions that systematically apply
6 progressive overload to not only build physical capacity but also strengthen EF under pressure. Such
7 research may offer a dual benefit: enhancing physical preparedness while fostering the cognitive
8 resilience required in professions where decision-making errors under stress can have critical
9 consequences.

Limitations

11 Several limitations of this review should be considered. First, although the included studies were
12 generally of high quality, with the majority employing true experimental designs, the quality of individual
13 studies was not formally appraised, nor was risk of bias assessed, as the review was designed to map the
14 literature rather than to evaluate study rigor. Second, only studies published in English were included,
15 which may have led to the exclusion of relevant findings published in other languages. Third, the broad
16 scope of the review, aimed at mapping the exercise–cognition literature incorporating progressive
17 overload, limited the depth of analysis. Consequently, the identified themes do not provide sufficient
18 granularity to support robust conclusions regarding each specific theme, nor do they allow for quantitative
19 synthesis such as pooled effect sizes.

Conclusion

21 This scoping review examined how progressive overload has been applied in long-term exercise
22 interventions targeting EF. Across the included studies, progressive overload—particularly when applied
23 to exercise intensity and task complexity—was consistently associated with improvements in EF, most
24 notably in working memory and inhibitory control. Positive effects were observed across diverse
25 populations, with the most studies focusing on children, adolescents, and older adults. However,
26 inconsistent reporting of progressive overload parameters and a lack of studies explicitly manipulating

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1 progressive overload as an independent variable represent key gaps in the literature. Additionally, few
2 studies have examined the application of progressively overloaded exercise interventions within
3 occupational settings where EF may be compromised due to stress, physical exertion, adverse
4 environmental conditions, or a combination of these factors. Furthermore, more research is needed to
5 isolate the effects of progressively overloaded resistance training on EF, as resistance-based interventions
6 were underrepresented in our review. For practitioners, incorporating progressive overload in both
7 physical and cognitive demands may optimize cognitive outcomes. For researchers, systematically
8 isolating and reporting progressive overload parameters offers an opportunity to refine and advance
9 exercise interventions designed to enhance EF.

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For Peer Review

Table 1
Characteristics of Sources of Evidence

Source	Population	Design	Intervention	Progressive Overload	EF	Findings
Abo-Zaid et al. (2024)	Children, 7–11 years, left hemiplegic CP (n=60)	Experimental	Frequency: 3x/week Intensity: 40-70% of HRM Time: 30min/session for 12 weeks Type: Cycling	Target: Intensity Frequency: 1x Magnitude: ~56%	IC	Large effect on IC ($\eta^2=0.45$)
Albinet et al. (2016)	Healthy sedentary adults, 60–75 years (n=36)	Experimental	Frequency: 2x/week Intensity: 40–65% HRR Time: 60min/session for 21 weeks Type: Aquaerobics and swimming	Target: Intensity Frequency: Unspecified Magnitude: ~63% over 21 weeks	IC, WM, CF	Moderate effect on IC ($\eta^2=0.088$) and WM ($\eta^2=0.08$)
Antunes et al. (2015)	Healthy sedentary elderly men, 60–75 years (n=46)	Experimental	Frequency: 3x/week Intensity: Ventilatory Threshold 1 Time: 20-60min/session for 6 months Type: Cycling	Target: Time Frequency: Unspecified Magnitude: ~200% over 6 months	WM	Positive effect on WM
Barcelos et al. (2015)	Older adults, mean age 82 (n=64)	Experimental	Frequency: 2-5x/week Intensity: Unspecified Time: 20-45min/session for 3 months Type: Exergaming	Target: Frequency, time Frequency: Unspecified Magnitude: Frequency: ~150% over 3 months; Time: ~125% over 3 months	IC, WM, CF	Large effect on IC ($\eta^2=0.46$)

Note. This table summarizes the characteristics of included studies examining the effects of exercise interventions on executive function (EF). Intervention parameters are structured according to the FITT principle: frequency (number of sessions per week), intensity (e.g., %HRM, %HRR, %VO_{2peak}), time (duration of each session and overall intervention), and type (mode of exercise). Progressive overload parameters are described as: target of progression (i.e., which FITT parameter is manipulated: frequency, intensity, time, or type), frequency of progression (e.g., every week, every two weeks), and magnitude of progression (e.g., 5%, 10%). Executive function (EF) domains include: IC = inhibitory control, WM = working memory, CF = cognitive flexibility. HRM = heart rate maximum; HRR = heart rate reserve; 1RM = 1-repetition maximum; VO_{2peak} = peak oxygen uptake; η^2 and η^2_p = (partial) eta squared effect sizes; d, g, and SMD = standardized effect size metrics (e.g., Cohen's d, Hedges' g, standardized mean difference); ADHD = attention deficit hyperactivity disorder; ASD = autism spectrum disorder; CP = cerebral palsy; FASD = fetal alcohol spectrum disorder.

Table 1 (continued)
Characteristics of Sources of Evidence

Source	Population	Design	Intervention	Progressive Overload	EF	Findings
Biino et al. (2021)	Preschool children, 3–6 years (n=36)	Quasi-experimental	Frequency: 2x/week Intensity: Unspecified Time: 45min/session for 12 weeks Type: Cognitively enriched physical activity or swimming	Target: Type Frequency: Unspecified Magnitude: Unspecified (Increasing difficulty)	IC, WM, CF	Large effect on WM ($\eta^2 = 0.53$)
Brydges et al. (2020)	Older women, 70–80 years, mild cognitive impairment (n=86)	Experimental	Frequency: 2x/week Intensity: 40–80% HRR Time: 60min/session for 6 months Type: Aerobic and resistance exercises	Target: Intensity Frequency: Unspecified Magnitude: ~100% over 12 weeks	WM, CF	Small to moderate effect on CF ($g = 0.27, 0.59$)
Byun et al. (2024)	Healthy adults, 55–80 years (n=81)	Experimental	Frequency: 3x/week Intensity: 35% of VO_{2peak} Time: 30–50min/session for 12 weeks Type: Cycling	Target: Time Frequency: 2x Magnitude: ~33%	IC	Positive effect on IC
Damirchi et al. (2018)	Older women, 60–85 years, mild cognitive impairment, (n=44)	Experimental	Frequency: 3x/week Intensity: 55–75% HRR Time: 30min/session for 8 weeks Type: Aerobic and mental training	Target: Intensity, time Frequency: Unspecified Magnitude: Intensity: ~36% over 8 weeks; Time: ~100% over 8 weeks	WM	No effect on WM

Note. This table summarizes the characteristics of included studies examining the effects of exercise interventions on executive function (EF). Intervention parameters are structured according to the FITT principle: frequency (number of sessions per week), intensity (e.g., %HRM, %HRR, % VO_{2peak}), time (duration of each session and overall intervention), and type (mode of exercise). Progressive overload parameters are described as: target of progression (i.e., which FITT parameter is manipulated: frequency, intensity, time, or type), frequency of progression (e.g., every week, every two weeks), and magnitude of progression (e.g., 5%, 10%). Executive function (EF) domains include: IC = inhibitory control, WM = working memory, CF = cognitive flexibility. HRM = heart rate maximum; HRR = heart rate reserve; 1RM = 1-repetition maximum; VO_{2peak} = peak oxygen uptake; η^2 and ηp^2 = (partial) eta squared effect sizes; d, g, and SMD = standardized effect size metrics (e.g., Cohen’s d, Hedges’ g, standardized mean difference); ADHD = attention deficit hyperactivity disorder; ASD = autism spectrum disorder; CP = cerebral palsy; FASD = fetal alcohol spectrum disorder.

Table 1 (continued)*Characteristics of Sources of Evidence*

Source	Population	Design	Intervention	Progressive Overload	EF	Findings
Ghaleni et al. (2024)	Older men, mean age 66 (n=36)	Experimental	Frequency: 2x/week Intensity: Moderate Time: 45min/session for 12 weeks Type: Aerobic exercise combined with amygdala and insula retraining	Target: Intensity, type Frequency: Unspecified Magnitude: Unspecified (Increasing difficulty)	WM	Large effect on WM ($\eta^2=0.44$)
Hou et al. (2023)	Older adults with hypertension, 60 to 80 years (n=128)	Experimental	Frequency: 3x/week Intensity: 60%-69% HRM Time: 60 min/session for 16 weeks Type: Exergaming or cycling	Target: Type Frequency: Unspecified Magnitude: Unspecified (Increasing difficulty)	IC, WM, CF	Moderate effect on WM for exergaming ($d=0.64$) and cycling ($d=0.72$)
Hu et al. (2024)	Obese junior high students (n=28) and normal-weight peers (n=24), 13-14 years	Quasi-experimental	Frequency: 4x/week Intensity: 60%-69% HRM Time: 40 min/session for 14 weeks Type: Aerobic and resistance exercise	Target: Type Frequency: Unspecified Magnitude: Unspecified (Increasing complexity)	IC, WM, CF	Moderate effect on IC ($\eta p^2 = 0.071$) and CF ($\eta p^2 = 0.073$)
Huang et al. (2024)	Typically developing children (n=17), and ADHD (n=38), 6-12 years	Experimental	Frequency: 2x/week Intensity: 64%-95% of HRM Time: 60min/session for 8 weeks Type: Rope skipping and sport games	Target: Type Frequency: 3x Magnitude: Unspecified (Increasing complexity)	WM	Moderate effect on WM ($d=0.595$)

Note. This table summarizes the characteristics of included studies examining the effects of exercise interventions on executive function (EF). Intervention parameters are structured according to the FITT principle: frequency (number of sessions per week), intensity (e.g., %HRM, %HRR, %VO₂peak), time (duration of each session and overall intervention), and type (mode of exercise). Progressive overload parameters are described as: target of progression (i.e., which FITT parameter is manipulated: frequency, intensity, time, or type), frequency of progression (e.g., every week, every two weeks), and magnitude of progression (e.g., 5%, 10%). Executive function (EF) domains include: IC = inhibitory control, WM = working memory, CF = cognitive flexibility. HRM = heart rate maximum; HRR = heart rate reserve; 1RM = 1-repetition maximum; VO₂peak = peak oxygen uptake; η^2 and ηp^2 = (partial) eta squared effect sizes; d, g, and SMD = standardized effect size metrics (e.g., Cohen's d, Hedges' g, standardized mean difference); ADHD = attention deficit hyperactivity disorder; ASD = autism spectrum disorder; CP = cerebral palsy; FASD = fetal alcohol spectrum disorder.

Table 1 (continued)
Characteristics of Sources of Evidence

Source	Population	Design	Intervention	Progressive Overload	EF	Findings
Iuliano et al. (2015)	Healthy adults, mean age 67 years (n=80)	Experimental	Frequency: 3x/week Intensity: Cardiovascular: 50-80% HRR; Resistance: 60-85% 1RM Time: 30min/session for 12 weeks Type: Resistance, cardiovascular, and postural exercise	Target: Intensity Frequency: Cardiovascular: 5x; Resistance: 2x Magnitude: Cardiovascular: ~8%; Resistance: ~13%	IC, CF	Positive effect on IC and CF (cardiovascular)
Jonasson et al. (2016)	Sedentary older adults, 64–78 years (n=58)	Experimental	Frequency: 3x/week Intensity: 40–80% HRM Time: 30–60min/session for 6 months Type: Aerobic exercise	Target: Intensity Frequency: Unspecified Magnitude: ~100% over 6 months	Overall EF	Positive effect on overall EF
Kleinloog et al. (2019)	Sedentary older men, 60-70 years (n=17)	Experimental	Frequency: 3x/week Intensity: 45%-70% max workload Time: 50 min/session for 8 weeks Type: Cycling	Target: Intensity Frequency: 3x Magnitude: Intensity adjusted to improvements	Overall EF	Positive effect on overall EF
Koutsandréou et al. (2016)	Elementary school children, 9–10 years (n=71)	Experimental	Frequency: 3x/week Intensity: 60-70% HRM for cardiovascular exercise; 55-65% for motor exercise Time: 45min/session for 10 weeks Type: Cardiovascular and motor exercise	Target: Type Frequency: Unspecified Magnitude: Unspecified (Increasing complexity)	WM	Large effect on WM ($\eta^2 = 0.267$)

Note. This table summarizes the characteristics of included studies examining the effects of exercise interventions on executive function (EF). Intervention parameters are structured according to the FITT principle: frequency (number of sessions per week), intensity (e.g., %HRM, %HRR, %VO₂peak), time (duration of each session and overall intervention), and type (mode of exercise). Progressive overload parameters are described as: target of progression (i.e., which FITT parameter is manipulated: frequency, intensity, time, or type), frequency of progression (e.g., every week, every two weeks), and magnitude of progression (e.g., 5%, 10%). Executive function (EF) domains include: IC = inhibitory control, WM = working memory, CF = cognitive flexibility. HRM = heart rate maximum; HRR = heart rate reserve; 1RM = 1-repetition maximum; VO₂peak = peak oxygen uptake; η^2 and ηp^2 = (partial) eta squared effect sizes; d, g, and SMD = standardized effect size metrics (e.g., Cohen’s d, Hedges’ g, standardized mean difference); ADHD = attention deficit hyperactivity disorder; ASD = autism spectrum disorder; CP = cerebral palsy; FASD = fetal alcohol spectrum disorder.

Table 1 (continued)*Characteristics of Sources of Evidence*

Source	Population	Design	Intervention	Progressive Overload	EF	Findings
Kubitz et al.(2023)	Overweight adults with sleep apnea, 30–65 years (n=15)	Pre-experimental	Frequency: 3x/week Intensity: 60–90% HRR; 60% 1RM Time: 60-65min/session for 6 weeks Type: Aerobic and resistance exercise	Target: Intensity Frequency: Unspecified Magnitude: Unspecified	IC	Large effect on IC with moderate-to-severe obstructive sleep apnea ($\eta^2=0.737$)
Law et al. (2018)	Older adults, 60–85 years, mild cognitive impairment (n=43)	Pre-experimental	Frequency: 1–2x/week Intensity: Moderate Time: 50 min/session for 10 weeks Type: Functional task exercise	Target: Type Frequency: Unspecified Magnitude: Unspecified (Increasing complexity)	Overall EF	Moderate effect on overall EF (d=0.774)
Leckie et al. (2014)	Healthy adults, 55–80 years (n=90)	Experimental	Frequency: Unspecified Intensity: 50–75% HRR Time: 10-40min/session for 12 months Type: Walking	Target: Intensity, time Frequency: Intensity: 1x; Time: 6x Magnitude: Intensity: ~23%; Time: ~27%	CF	Positive effect on CF
Lee et al., (2017)	Boys with ADHD, mean age 9 years (n=12)	Experimental	Frequency: 3x/week Intensity: 45–75% HRR Time: 60min/session for 12 weeks Type: Fundamental movement skills	Target: Intensity Frequency: 2x Magnitude: ~18%	IC	Positive effect on IC

Note. This table summarizes the characteristics of included studies examining the effects of exercise interventions on executive function (EF). Intervention parameters are structured according to the FITT principle: frequency (number of sessions per week), intensity (e.g., %HRM, %HRR, %VO₂peak), time (duration of each session and overall intervention), and type (mode of exercise). Progressive overload parameters are described as: target of progression (i.e., which FITT parameter is manipulated: frequency, intensity, time, or type), frequency of progression (e.g., every week, every two weeks), and magnitude of progression (e.g., 5%, 10%). Executive function (EF) domains include: IC = inhibitory control, WM = working memory, CF = cognitive flexibility. HRM = heart rate maximum; HRR = heart rate reserve; 1RM = 1-repetition maximum; VO₂peak = peak oxygen uptake; η^2 and ηp^2 = (partial) eta squared effect sizes; d, g, and SMD = standardized effect size metrics (e.g., Cohen's d, Hedges' g, standardized mean difference); ADHD = attention deficit hyperactivity disorder; ASD = autism spectrum disorder; CP = cerebral palsy; FASD = fetal alcohol spectrum disorder.

Table 1 (continued)
Characteristics of Sources of Evidence

Source	Population	Design	Intervention	Progressive Overload	EF	Findings
Liu et al. (2022)	Older adults, 50–85 years, Parkinson’s disease, (n=28)	Experimental	Frequency: 2x/week Intensity: Unspecified Time: 60 min/session for 8 weeks Type: Square-stepping exercise	Target: Type Frequency: Unspecified Magnitude: Unspecified (Increasing complexity)	WM, CF	Positive effect on WM
Luo et al. (2022)	Older adults, 55–80 years, mild cognitive impairment (n=48)	Experimental	Frequency: 3x/week Intensity: Unspecified Time: 60min/session for 40 weeks Type: Wuqinxi exercise	Target: Type Frequency: 1x Magnitude: Unspecified (Increasing complexity)	WM	Positive effect on WM
Meha et al. (2024)	Young football players, 13–19 years (n=1253 individuals, 55 teams)	Experimental	Frequency: 2x/week Intensity: Unspecified Time: 15–20 min/session for 9 months Type: Injury prevention program (FUNBALL) with integrated cognitive challenges	Target: Type Frequency: 5-6x Magnitude: Unspecified (Increasing complexity)	WM, IC	Small to moderate effect on WM (d=0.167, 0.610, 0.276) and small effect on CF (d=0.438)
Mora-Gonzalez et al. (2023)	Obese children, mean age 10 (n=67)	Experimental	Frequency: 3–5x/week Intensity: Moderate to vigorous Time: 90min/session for 20 weeks Type: Combined aerobic and resistance exercises	Target: Intensity Frequency: Unspecified Magnitude: Intensity adjusted to improvements	IC, WM	Positive effect on brain activation during WM and IC tasks

Note. This table summarizes the characteristics of included studies examining the effects of exercise interventions on executive function (EF). Intervention parameters are structured according to the FITT principle: frequency (number of sessions per week), intensity (e.g., %HRM, %HRR, %VO₂peak), time (duration of each session and overall intervention), and type (mode of exercise). Progressive overload parameters are described as: target of progression (i.e., which FITT parameter is manipulated: frequency, intensity, time, or type), frequency of progression (e.g., every week, every two weeks), and magnitude of progression (e.g., 5%, 10%). Executive function (EF) domains include: IC = inhibitory control, WM = working memory, CF = cognitive flexibility. HRM = heart rate maximum; HRR = heart rate reserve; 1RM = 1-repetition maximum; VO₂peak = peak oxygen uptake; η^2 and ηp^2 = (partial) eta squared effect sizes; d, g, and SMD = standardized effect size metrics (e.g., Cohen’s d, Hedges’ g, standardized mean difference); ADHD = attention deficit hyperactivity disorder; ASD = autism spectrum disorder; CP = cerebral palsy; FASD = fetal alcohol spectrum disorder.

Table 1 (continued)*Characteristics of Sources of Evidence*

Source	Population	Design	Intervention	Progressive Overload	EF	Findings
Norouzi et al. (2019)	Healthy adults, mean age 68 years (n=60)	Experimental	Frequency: 3x/week Intensity: Varied Time: 60–80min/session for 4 weeks Type: Motor-cognitive dual-task training and motor-motor dual-task training	Target: Intensity Frequency: 2x Magnitude: ~23%	WM	Large effect on WM ($\eta^2 = 0.38$)
Peven et al., (2020)	Overweight and obese adults, 20–67 years (n=125)	Experimental	Frequency: ~5x/week Intensity: Moderate to high Time: 100-250min/week for 12 months Type: Aerobic exercise combined with dietary intervention	Target: Time Frequency: Unspecified Magnitude: ~50% over 9 weeks (moderate exercise group), ~150% over 25 weeks (high exercise group)	IC, WM, CF	No effect on IC, WM, CF
Pritchard Orr et al. (2018)	Children with FASD, 7–14 years (n=30)	Quasi-experimental	Frequency: 2x/week Intensity: Unspecified Time: 90min/session for 8 weeks Type: Motor skill training and games	Target: Type Frequency: Unspecified Magnitude: Unspecified (Increasing complexity)	Overall EF	Small effect on overall EF ($d=0.46$)
Roig-Coll et al. (2020)	Healthy adults, 50–70 years (n=96)	Experimental	Frequency: 5x/week Intensity: Light to moderate Time: 30–45min/session for 12 weeks Type: Walking	Target: Intensity, time Frequency: 1x Magnitude: Intensity: ~37%; Time: ~50%	IC, WM, CF	Small effect on CF (SMD = -0.33) and WM (SMD = 0.29)

Note. This table summarizes the characteristics of included studies examining the effects of exercise interventions on executive function (EF). Intervention parameters are structured according to the FITT principle: frequency (number of sessions per week), intensity (e.g., %HRM, %HRR, %VO₂peak), time (duration of each session and overall intervention), and type (mode of exercise). Progressive overload parameters are described as: target of progression (i.e., which FITT parameter is manipulated: frequency, intensity, time, or type), frequency of progression (e.g., every week, every two weeks), and magnitude of progression (e.g., 5%, 10%). Executive function (EF) domains include: IC = inhibitory control, WM = working memory, CF = cognitive flexibility. HRM = heart rate maximum; HRR = heart rate reserve; 1RM = 1-repetition maximum; VO₂peak = peak oxygen uptake; η^2 and ηp^2 = (partial) eta squared effect sizes; d, g, and SMD = standardized effect size metrics (e.g., Cohen's d, Hedges' g, standardized mean difference); ADHD = attention deficit hyperactivity disorder; ASD = autism spectrum disorder; CP = cerebral palsy; FASD = fetal alcohol spectrum disorder.

Table 1 (continued)
Characteristics of Sources of Evidence

Source	Population	Design	Intervention	Progressive Overload	EF	Findings
Şendil et al. (2024)	Preschool children, 5–6 years (n=41)	Quasi-experimental	Frequency: 2x/week Intensity: Low to moderate Time: 30min/session for 8 weeks Type: Structured coordinative exercises focusing on motor skills	Target: Time, type Frequency: 7x Magnitude: Time: ~5%; Type: Unspecified (Increasing complexity)	IC	No effect on IC
Sexton et al. (2020)	Healthy adults, 60–85 years (n=46)	Experimental	Frequency: 3x/week Intensity: 55–85% HRM Time: 30min/session for 12 weeks Type: Cycling	Target: Intensity Frequency: Unspecified Magnitude: ~55% over 12 weeks	WM, CF	No effect on WM and CF
Shen et al. (2020)	Preschool children, 4–5 years (n=60)	Experimental	Frequency: 3x/week Intensity: Moderate Time: 40–50min/session for 8 weeks Type: Street-dancing	Target: Type Frequency: 1x Magnitude: Unspecified (Increasing complexity)	IC, WM, CF	Moderate effect on IC ($\eta^2 = 0.108$), WM ($\eta^2 = 0.074$) and CF ($\eta^2 = 0.087$)
Sun et al. (2024)	Children with ADHD, 8–13 years (n=49)	Experimental	Frequency: 2x/week Intensity: Unspecified Time: 30 min/session for 8 weeks Type: Game-based HIIT and structured aerobic exercise	Target: Intensity Frequency: 3x Magnitude: Unspecified	IC, WM, CF	No effect on objectively measured IC, WM, CF

Note. This table summarizes the characteristics of included studies examining the effects of exercise interventions on executive function (EF). Intervention parameters are structured according to the FITT principle: frequency (number of sessions per week), intensity (e.g., %HRM, %HRR, %VO₂peak), time (duration of each session and overall intervention), and type (mode of exercise). Progressive overload parameters are described as: target of progression (i.e., which FITT parameter is manipulated: frequency, intensity, time, or type), frequency of progression (e.g., every week, every two weeks), and magnitude of progression (e.g., 5%, 10%). Executive function (EF) domains include: IC = inhibitory control, WM = working memory, CF = cognitive flexibility. HRM = heart rate maximum; HRR = heart rate reserve; 1RM = 1-repetition maximum; VO₂peak = peak oxygen uptake; η^2 and ηp^2 = (partial) eta squared effect sizes; d, g, and SMD = standardized effect size metrics (e.g., Cohen’s d, Hedges’ g, standardized mean difference); ADHD = attention deficit hyperactivity disorder; ASD = autism spectrum disorder; CP = cerebral palsy; FASD = fetal alcohol spectrum disorder.

Table 1 (continued)*Characteristics of Sources of Evidence*

Source	Population	Design	Intervention	Progressive Overload	EF	Findings
Tabei et al., (2023)	Healthy adults, mean age 71 years (n=114)	Quasi-experimental	Frequency: 1x/week Intensity: Unspecified Time: 60 min/session for 6 months Type: Exercise with music	Target: Intensity Frequency: 26x Magnitude: Unspecified	WM	Positive effect on WM
Tarumi et al. (2022)	Healthy adults, 60–80 years (n=73)	Experimental	Frequency: 3-5x/week Intensity: 75–90% HRM Time: 25–40min/session for 12 months Type: Aerobic exercise	Target: Time, frequency, intensity Frequency: Time: 2x; Frequency: 2x; Intensity: 1x Magnitude: Time: ~27%; Frequency: ~67%; Intensity: ~9%	Cognitive composite score, including WM	Positive effect on cognitive composite score
Tsai et al. (2017)	Healthy elderly men, 60–80 years (n=64)	Experimental	Frequency: 3x/week Intensity: 50–75% HRR Time: 40min/session for 24 weeks Type: Open- and closed-skill exercise	Target: Intensity, type Frequency: 1x Magnitude: Intensity: ~32%; Type: Unspecified (Increasing complexity)	WM, CF	Large effect on WM ($\eta^2 = 0.39$) and moderate effect on CF ($\eta^2 = 0.11$)
Venckunas et al. (2016)	Dinghy sailors, 15–18 years (n=8)	Quasi-experimental	Frequency: 3x/week Intensity: 95-145% of average 2000m pre-test running velocity Time: 37-106min/session for 7 weeks Type: Interval running	Target: Time Frequency: 6x Magnitude: ~25%	WM, CF	Positive effect on CF

Note. This table summarizes the characteristics of included studies examining the effects of exercise interventions on executive function (EF). Intervention parameters are structured according to the FITT principle: frequency (number of sessions per week), intensity (e.g., %HRM, %HRR, %VO₂peak), time (duration of each session and overall intervention), and type (mode of exercise). Progressive overload parameters are described as: target of progression (i.e., which FITT parameter is manipulated: frequency, intensity, time, or type), frequency of progression (e.g., every week, every two weeks), and magnitude of progression (e.g., 5%, 10%). Executive function (EF) domains include: IC = inhibitory control, WM = working memory, CF = cognitive flexibility. HRM = heart rate maximum; HRR = heart rate reserve; 1RM = 1-repetition maximum; VO₂peak = peak oxygen uptake; η^2 and ηp^2 = (partial) eta squared effect sizes; d, g, and SMD = standardized effect size metrics (e.g., Cohen's d, Hedges' g, standardized mean difference); ADHD = attention deficit hyperactivity disorder; ASD = autism spectrum disorder; CP = cerebral palsy; FASD = fetal alcohol spectrum disorder.

Table 1 (continued)
Characteristics of Sources of Evidence

Source	Population	Design	Intervention	Progressive Overload	EF	Findings
Vidoni et al. (2015)	Older adults, mean age 73 years (n=101)	Experimental	Frequency: 3–5x/week Intensity: 40–75% HRR Time: 12-75min/session for 26 weeks Type: Aerobic exercise	Target: Intensity, time Frequency: Intensity: 2x; Time: Weekly until goal achieved Magnitude: Intensity: ~19%; Time: 21min/week until goal achieved	IC, WM, CF	Positive effect for IC
Vieira et al. (2018)	Adults with coronary artery disease, 40–75 years (n=33)	Experimental	Frequency: 3x/week Intensity: 65–70% HRR Time: 71-86 min/session for 6 months Type: Virtual reality exercise	Target: Intensity, time Frequency: 1x Magnitude: Intensity: ~8%; Time: ~100%	IC, WM, CF	Positive effect on IC
Wang et al. (2017)	Adults with methamphetamine dependencies, 18–40 years (n=62)	Experimental	Frequency: 3x/week Intensity: 65-75% HRM Time: 30 min/session for 12 weeks Type: Aerobic exercise	Target: Intensity Frequency: 1x Magnitude: ~8%	IC	Positive effect on IC
Wang et al. (2018)	Older adults, mean age 71 years (n=27)	Quasi-experimental	Frequency: 3x/week Intensity: Resistance: 50–80% 1RM; Endurance: 70–75% HRM Time: 60 min/session for 12 weeks Type: Multicomponent exercise	Target: Intensity Frequency: Unspecified Magnitude: Resistance: ~55%; Endurance: maintained	Overall EF	Positive effect on overall EF

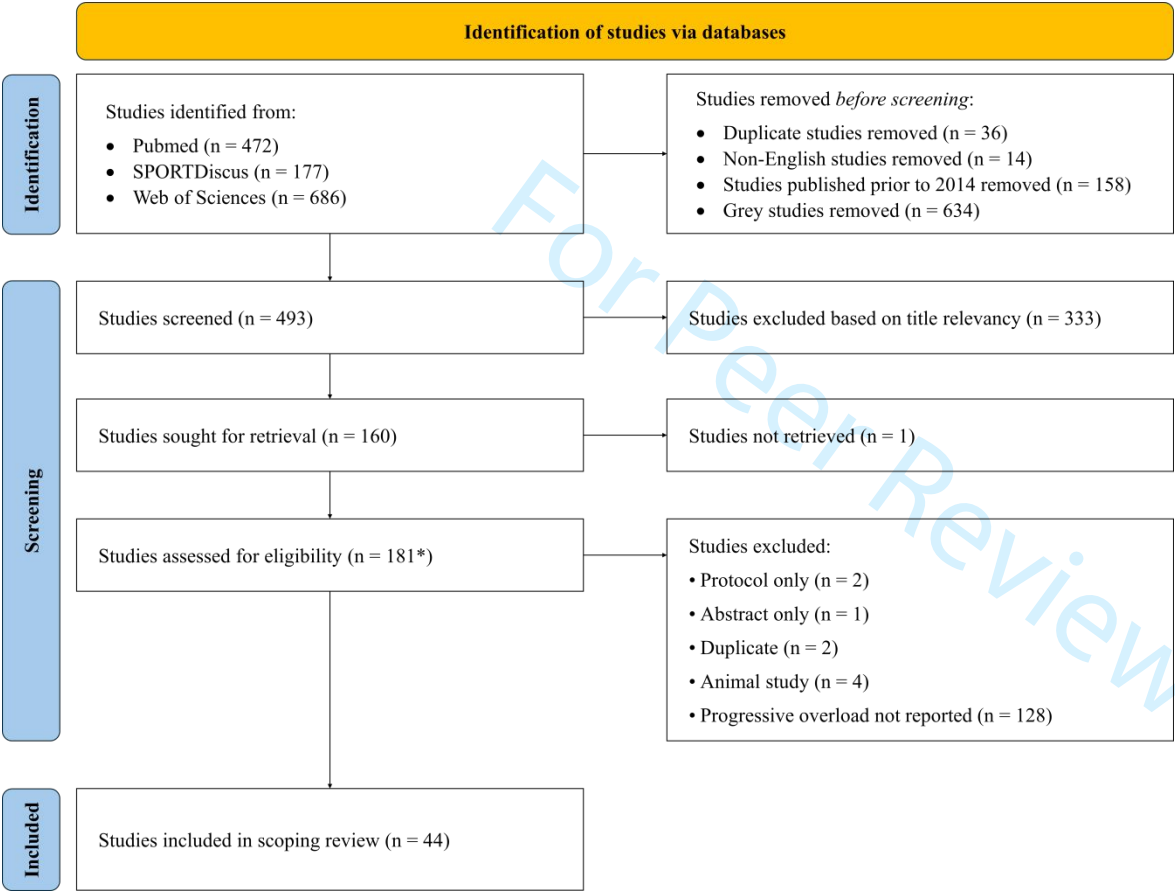
Note. This table summarizes the characteristics of included studies examining the effects of exercise interventions on executive function (EF). Intervention parameters are structured according to the FITT principle: frequency (number of sessions per week), intensity (e.g., %HRM, %HRR, %VO₂peak), time (duration of each session and overall intervention), and type (mode of exercise). Progressive overload parameters are described as: target of progression (i.e., which FITT parameter is manipulated: frequency, intensity, time, or type), frequency of progression (e.g., every week, every two weeks), and magnitude of progression (e.g., 5%, 10%). Executive function (EF) domains include: IC = inhibitory control, WM = working memory, CF = cognitive flexibility. HRM = heart rate maximum; HRR = heart rate reserve; 1RM = 1-repetition maximum; VO₂peak = peak oxygen uptake; η^2 and ηp^2 = (partial) eta squared effect sizes; d, g, and SMD = standardized effect size metrics (e.g., Cohen’s d, Hedges’ g, standardized mean difference); ADHD = attention deficit hyperactivity disorder; ASD = autism spectrum disorder; CP = cerebral palsy; FASD = fetal alcohol spectrum disorder.

Table 1 (continued)*Characteristics of Sources of Evidence*

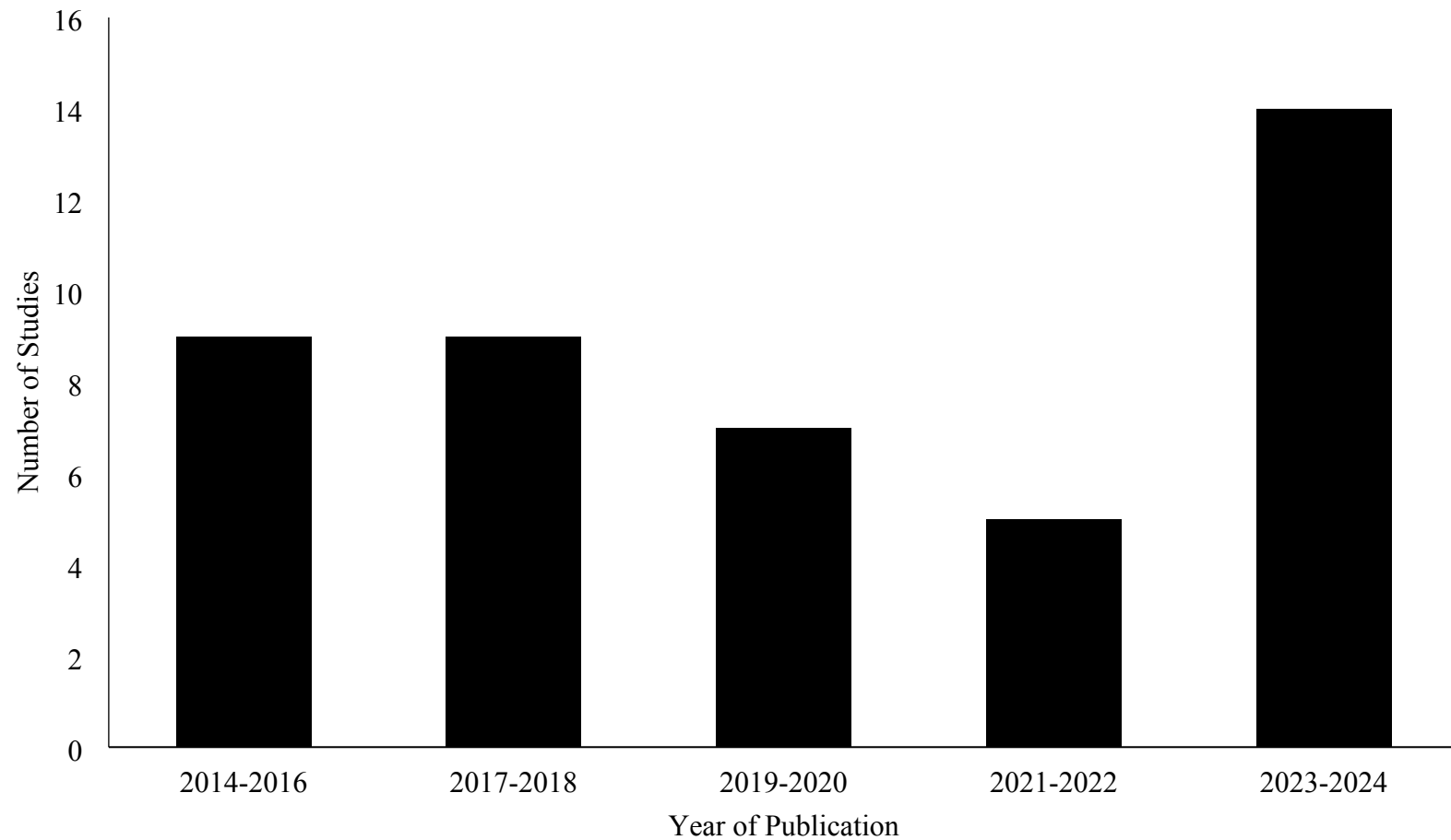
Source	Population	Design	Intervention	Progressive Overload	EF	Findings
Ye et al. (2024)	Adults, 45+ (n=2881)	Systematic review and meta-analysis	Frequency: 3-7x/week Intensity: Varied Time: 20–60 min/session for 48 weeks Type: Aerobic exercise	Target: Intensity Frequency: Varied Magnitude: Varied	IC, WM, CF	Small effect for progressive intensity interventions on IC ($g = 0.323$), WM ($g = 0.234$), and CF ($g = 0.335$)
Yu et al. (2018)	Older adults with Alzheimer's, mean age 78 years (n=28)	Pre-experimental	Frequency: 3x/week Intensity: 65–75% HRR Time: 25–55 min/session for 6 months Type: Cycling	Target: Time Frequency: Unspecified Magnitude: ~15%	Overall EF	No effect, suggesting that intervention limited decline in overall EF
Zhang et al. (2022)	Preschool children, 4–5 years (n=126)	Quasi-experimental	Frequency: 3x/week Intensity: 60-69% HRM Time: 40min/session for 12 weeks Type: Cognitive-motor games	Target: Type Frequency: 3x Magnitude: Unspecified (Increasing complexity)	WM	Moderate effect on WM ($\eta^2 = 0.107$)
Zhao et al. (2024)	Children with ASD, 9–12 years (n=30)	Experimental	Frequency: 4x/week Intensity: 60–69% HRM Time: 60–70 min/session for 12 weeks Type: Aquatic exercise	Target: Type Frequency: 1x Magnitude: Unspecified (Increasing complexity)	IC, CF, WM	Large effect on IC ($\eta^2 = 0.544$) and CF ($\eta^2 = 0.464, 0.542$)

Note. This table summarizes the characteristics of included studies examining the effects of exercise interventions on executive function (EF). Intervention parameters are structured according to the FITT principle: frequency (number of sessions per week), intensity (e.g., %HRM, %HRR, %VO₂peak), time (duration of each session and overall intervention), and type (mode of exercise). Progressive overload parameters are described as: target of progression (i.e., which FITT parameter is manipulated: frequency, intensity, time, or type), frequency of progression (e.g., every week, every two weeks), and magnitude of progression (e.g., 5%, 10%). Executive function (EF) domains include: IC = inhibitory control, WM = working memory, CF = cognitive flexibility. HRM = heart rate maximum; HRR = heart rate reserve; 1RM = 1-repetition maximum; VO₂peak = peak oxygen uptake; η^2 and ηp^2 = (partial) eta squared effect sizes; d, g, and SMD = standardized effect size metrics (e.g., Cohen's d, Hedges' g, standardized mean difference); ADHD = attention deficit hyperactivity disorder; ASD = autism spectrum disorder; CP = cerebral palsy; FASD = fetal alcohol spectrum disorder.

Figure 1
Selection of Sources of Evidence



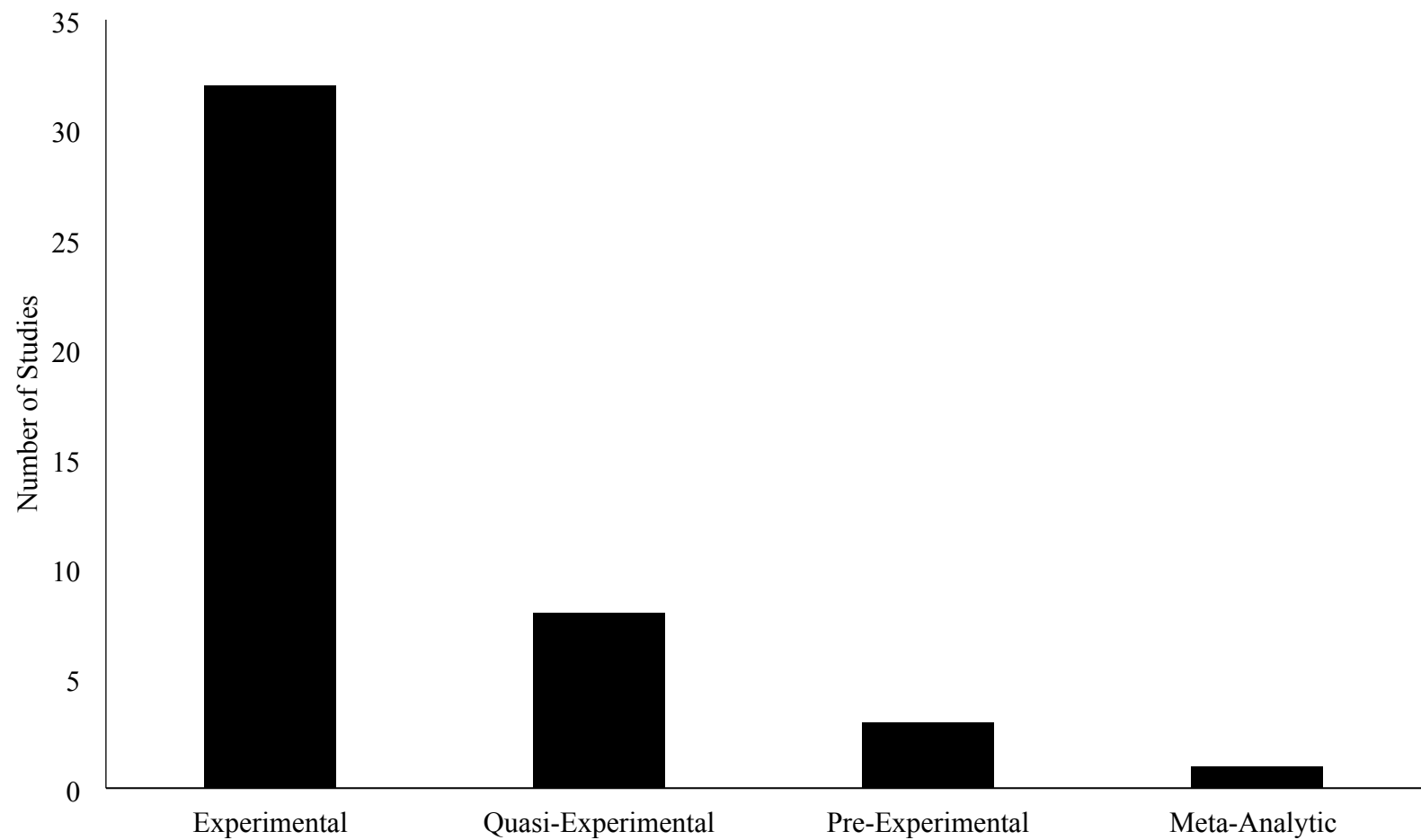
Note. PRISMA-ScR flow diagram illustrating the identification, screening, and inclusion process for sources of evidence included in the scoping review. * During the eligibility assessment, an additional 22 studies were identified from reference lists and analyzed, bringing the total number of studies assessed for eligibility to 181.

Figure 2*Number of Studies by Year of Publication*

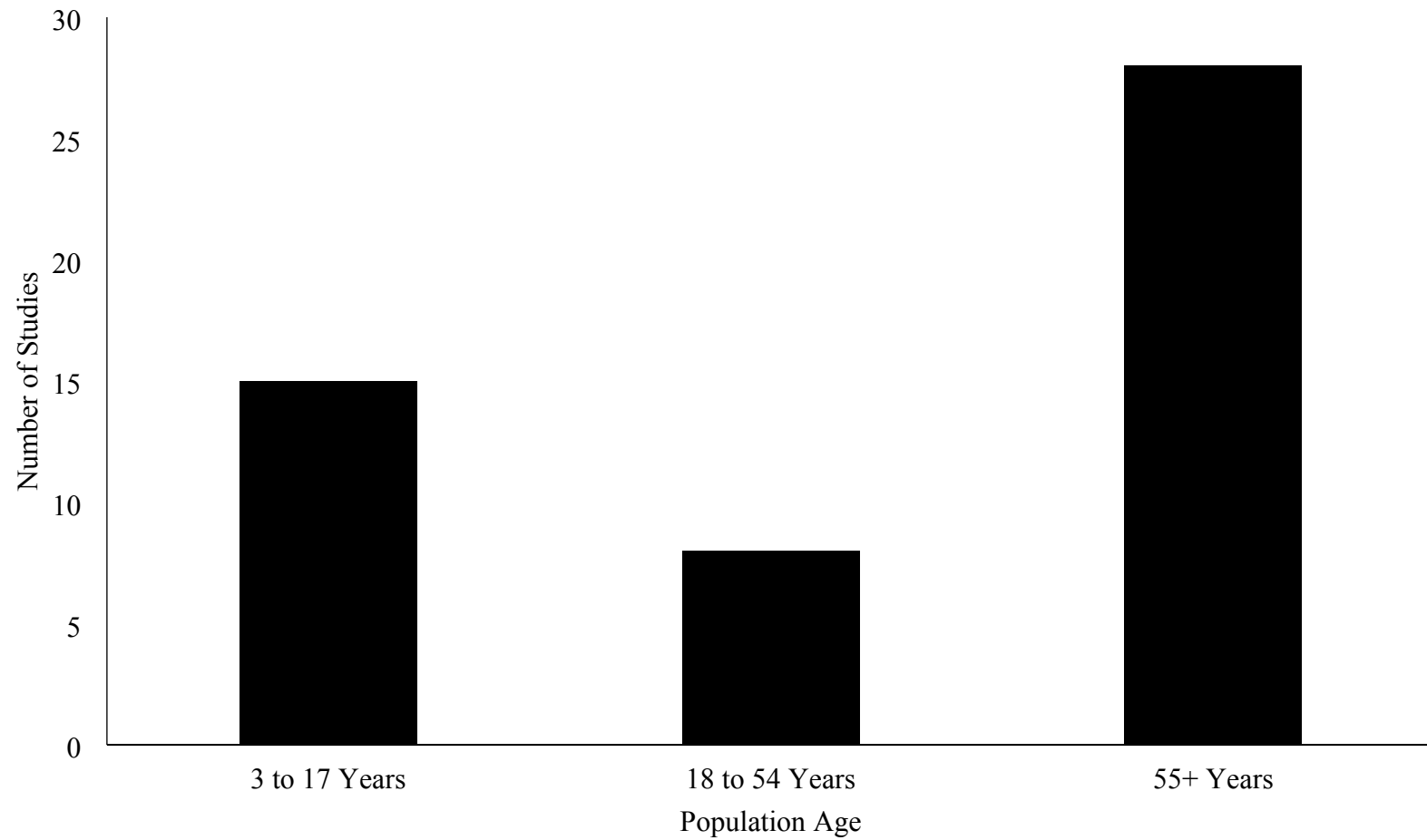
Note. Number of studies published between January 2014 and December 2024, grouped in two-year increments. The first interval spans three years (2014–2016).

Figure 3

Number of Studies by Design



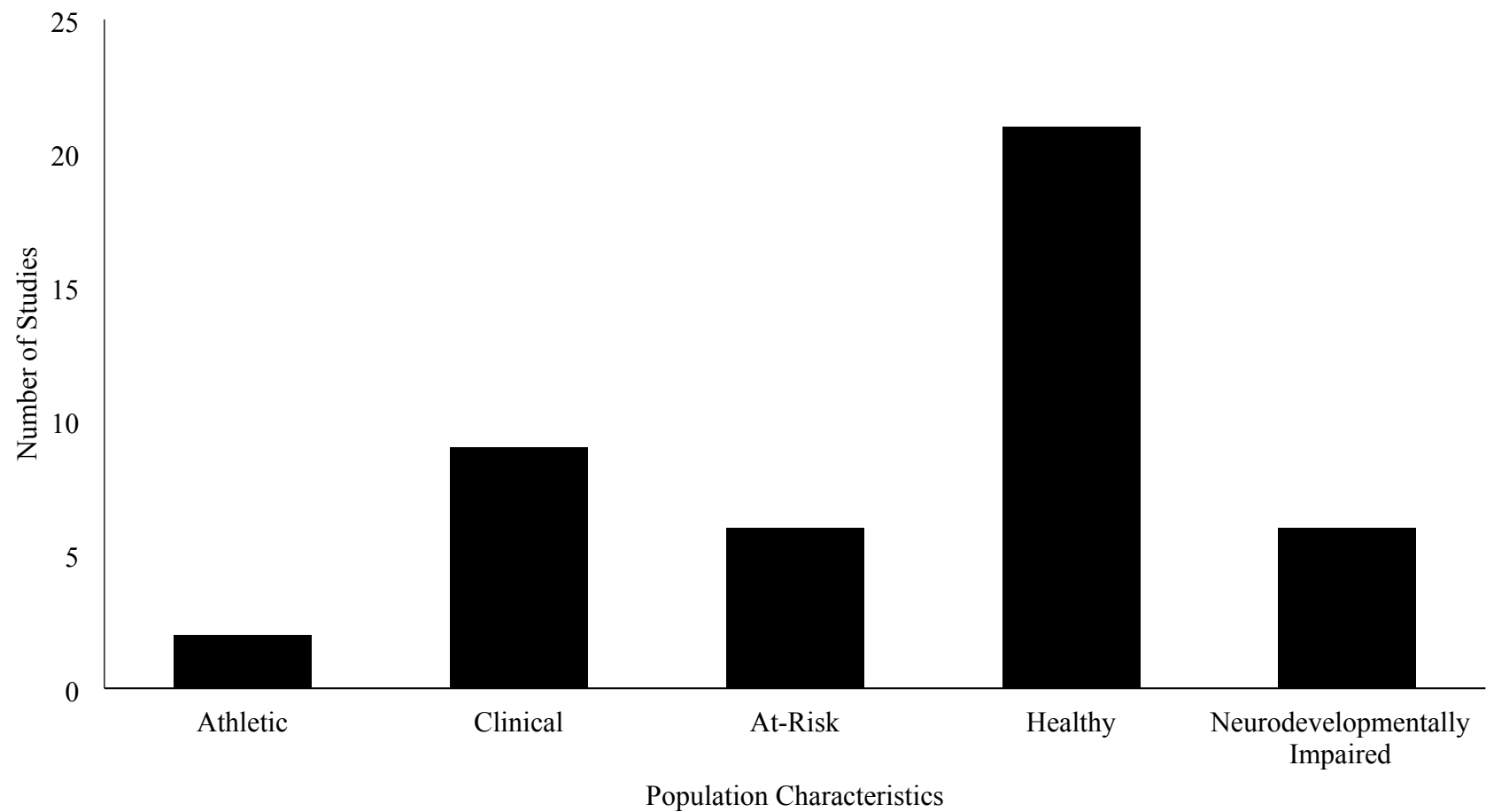
Note. Number of included studies by research design: experimental, quasi-experimental, pre-experimental, and meta-analytic.

Figure 4*Number of Studies by Population Age*

Note. Number of studies by age group: children/youth (3–17), adults (18–54), and older adults (55+). Totals exceed 44 due to overlap across groups.

Figure 5

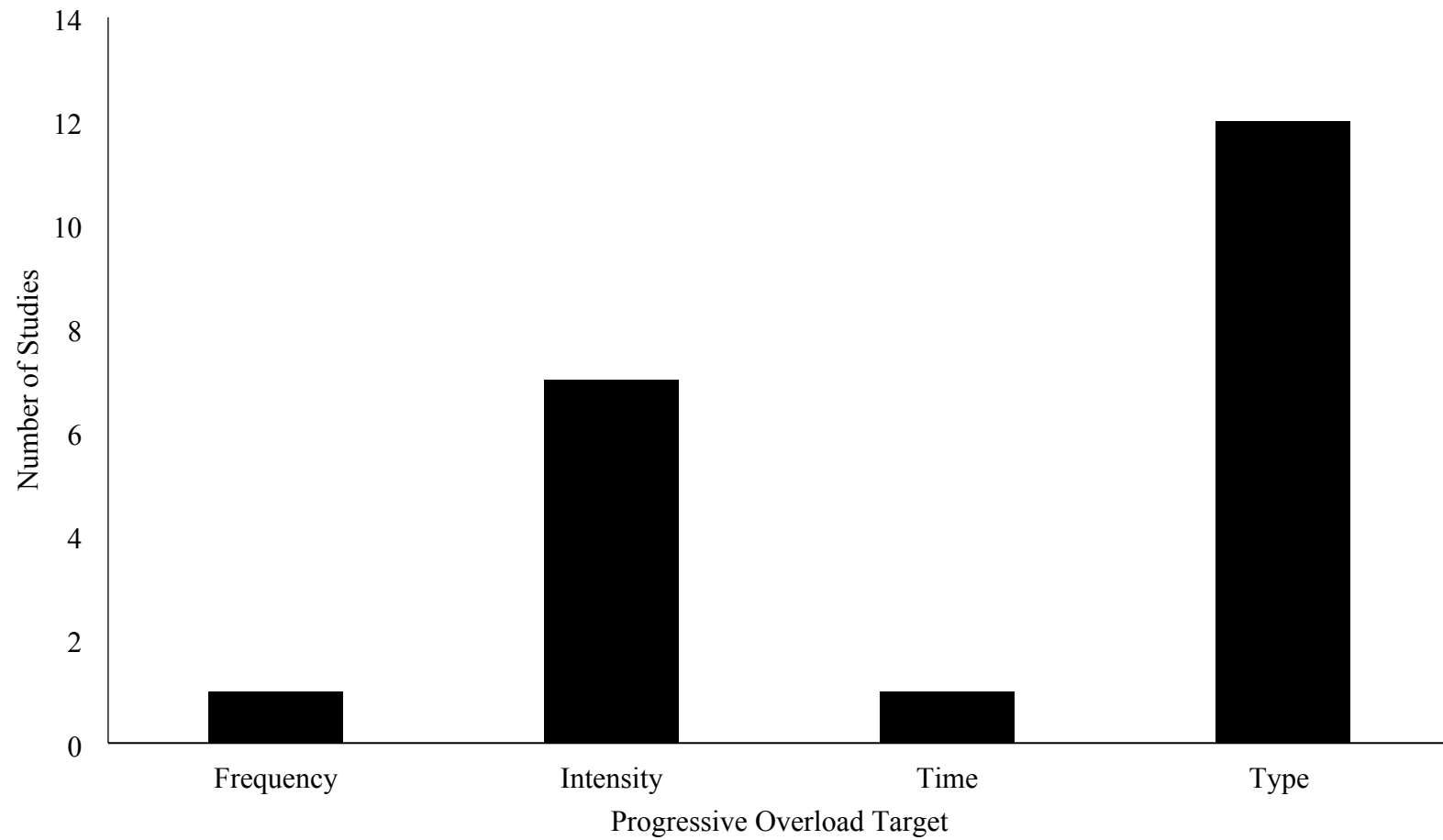
Number of Studies by Population Characteristics



Note. Number of studies by population characteristics: athletic, clinical, at-risk, healthy, and neurodevelopmentally impaired.

Figure 6

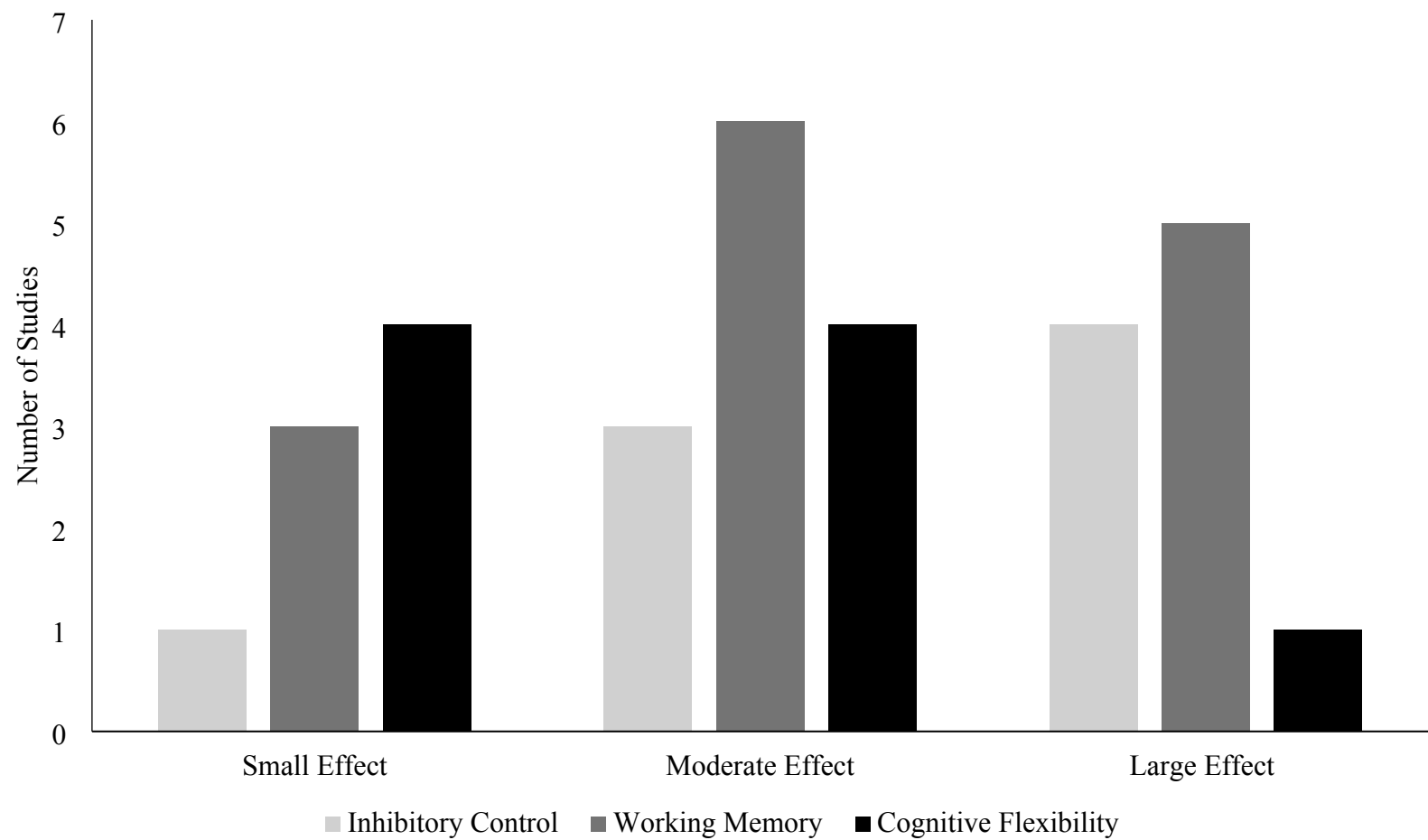
Number of Studies by Progressive Overload Parameter and Effect Size



Note. Distribution of progressive overload targets among studies reporting moderate to large effect sizes.

Figure 7

Number of Studies by Executive Function Domain and Effect Size



Note. Distribution of reported effect sizes (small, moderate, large) by executive function domain targeted in intervention studies.